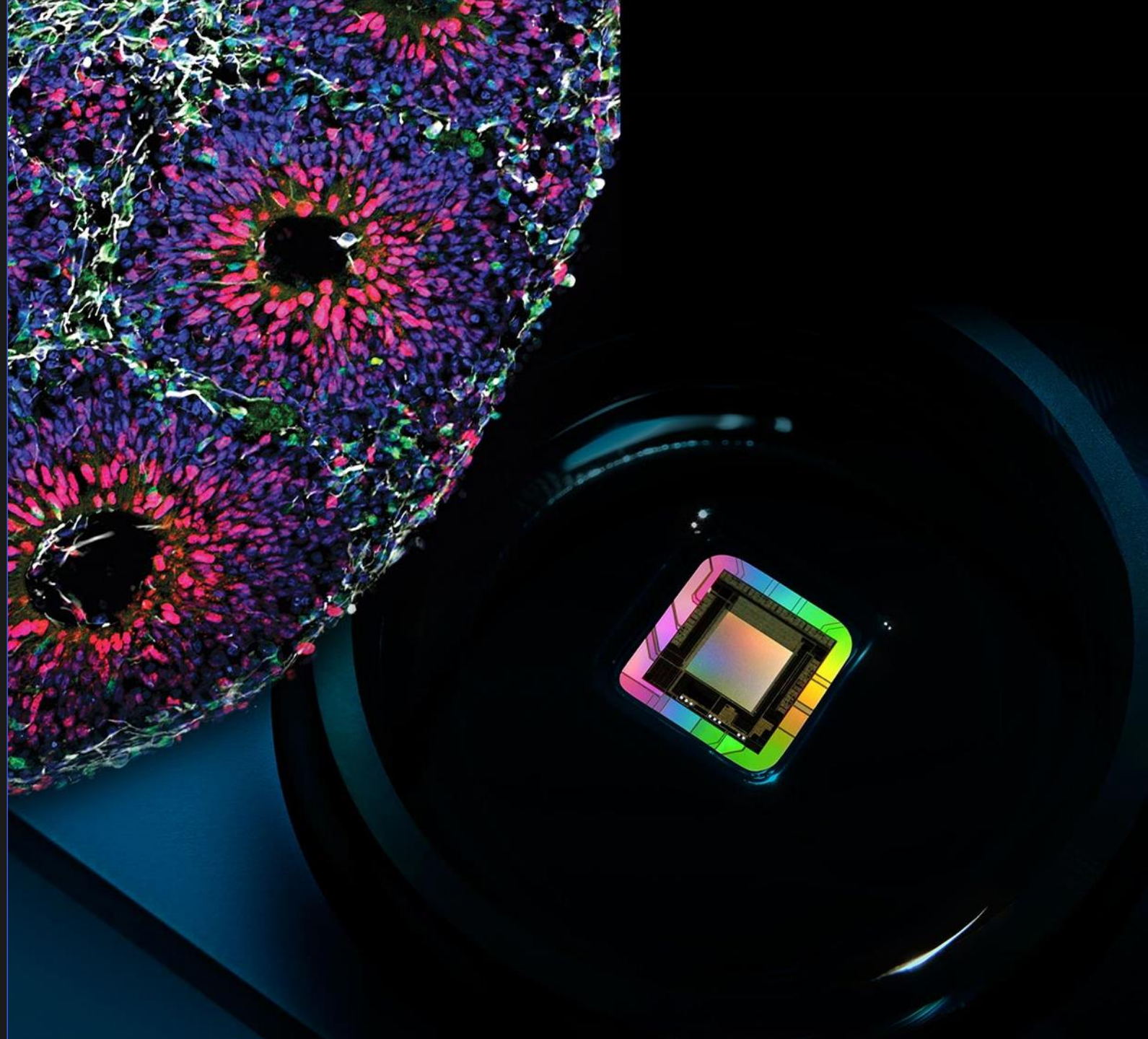




Microchip-based technologies for life science

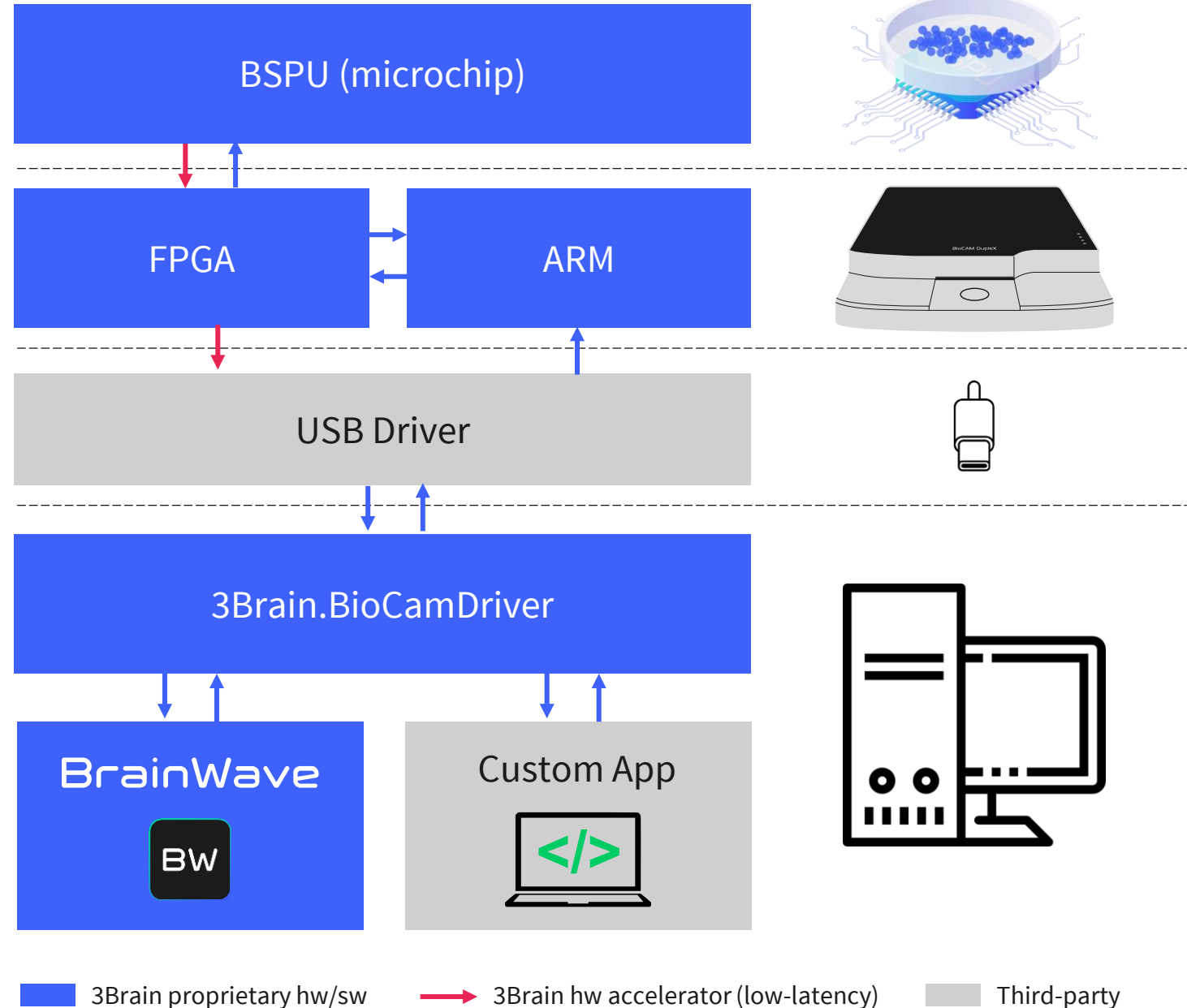
API v. 2.6 for BioCAM DupleX



Architecture and Specs

Specs

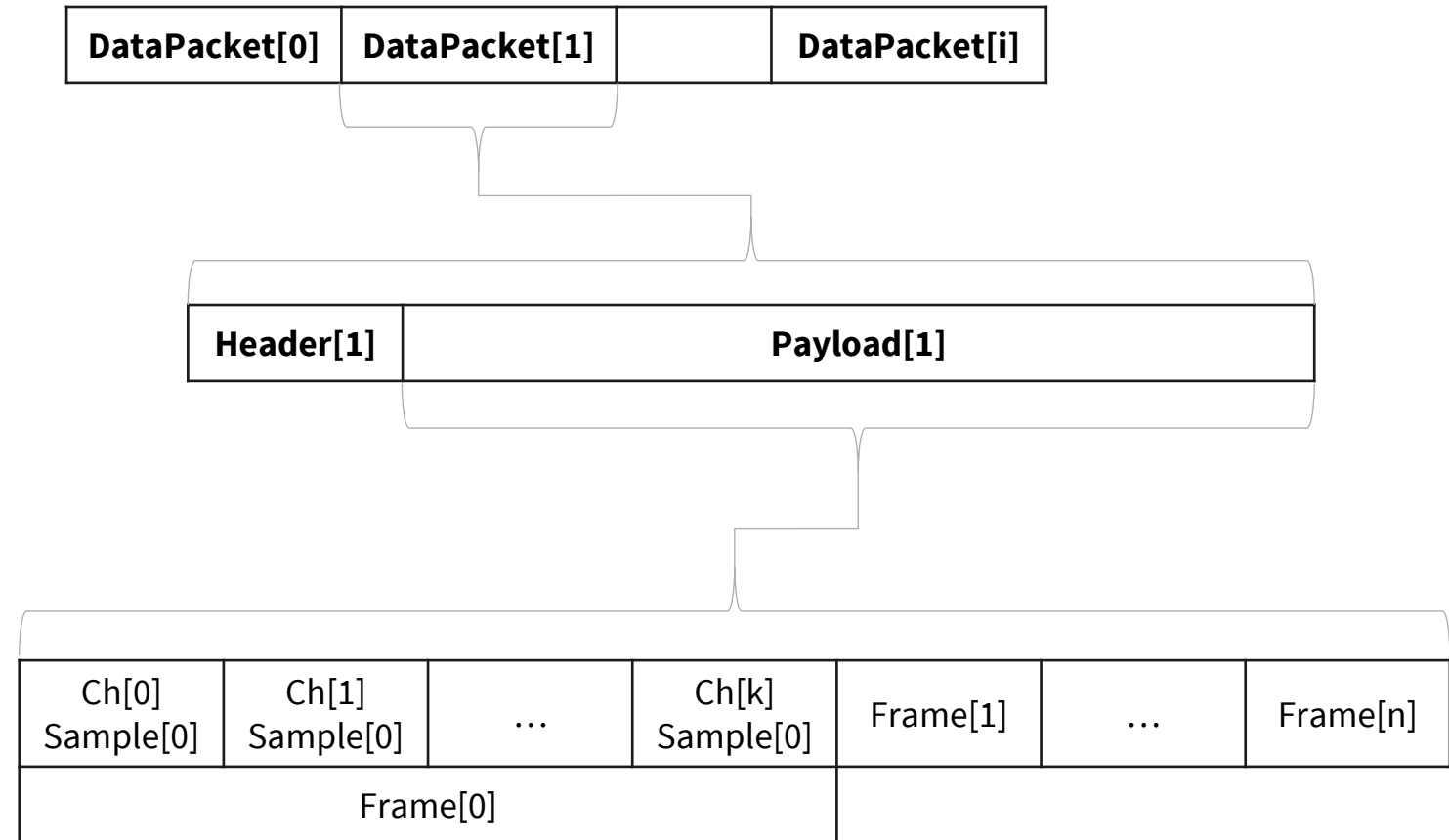
- Built for .NET Framework 4.7.
- Compatible with Windows 10 or 11.
- Can be used in C++, Java, Python, etc. by using a proper wrapper (not provided).
- Same toolset used in BrainWave (starting from BrainWave v. 5.0).



Reading Capabilities

Specs and Format

- Low-latency thanks to the 3Brain hw accelerator (see next slides).
- Data is received as DataPackets, with each DataPacket beign made up by a small Header, which contains useful information as the TimeStamp of the data, and the Payload, i.e. the data itself
- The size of the Payload is fixed during a certain acquisition and depends on the Acquisition Time Period (see next slides).
- Data inside the Payload is in raw format, as a sequence of frames where each frame contains the samples from all the electrodes.



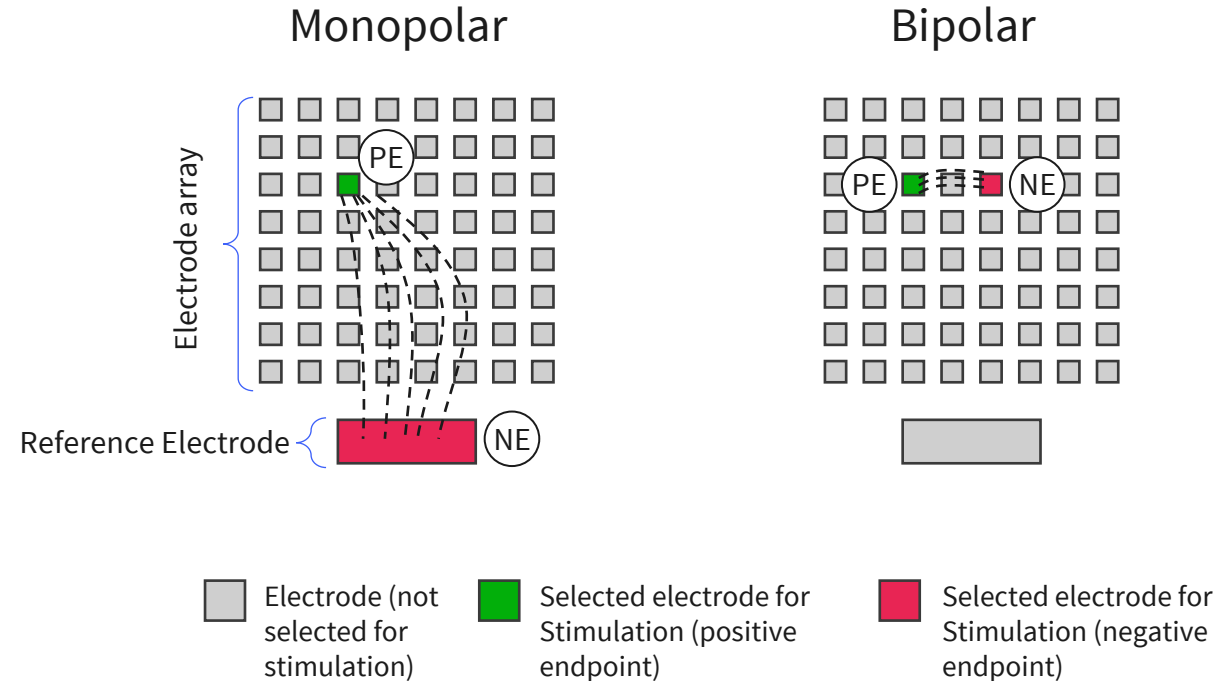
i = number of transmitted data packets; k = number of channels (electrodes); n = number of acquired frames

Stimulation Capabilities - 1

Types of Stimulation

Electrical stimuli are released between positive endpoints (PEs) and negative endpoints (NEs). At a given time, PEs and NEs can be assigned to either (and consequently stimulus released through):

1. one electrode (or multiple electrodes) on the array and the reference electrode (Monopolar stimulation).
2. two electrodes or groups of electrodes on the array (Bipolar stimulation).

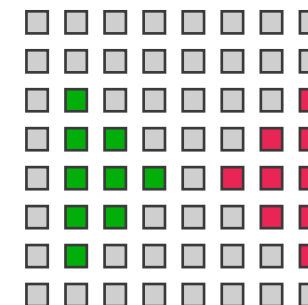


Spatial Configurations (SCs)

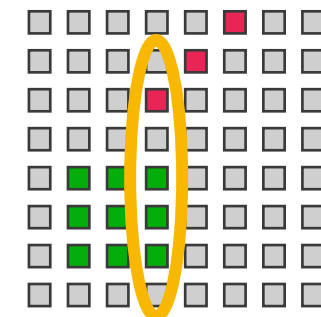
A Spatial Configuration (SC) is the positive and negative endpoints that are selected for the stimulation *at a given time*. Multiple and different SCs can quickly follow one another (see next slide). SCs must meet these conditions:

1. The combined number of endpoints is $\leq 1'000$ (and th.
2. Positive and negative endpoints do not share the same column(s) in the electrode array. In the aside figure, SC₁ is allowed, while SC₂ is not because one negative endpoint shares an electrode column that is already taken by 3 positive endpoints.

SC₁ (allowed)



SC₂ (not allowed)



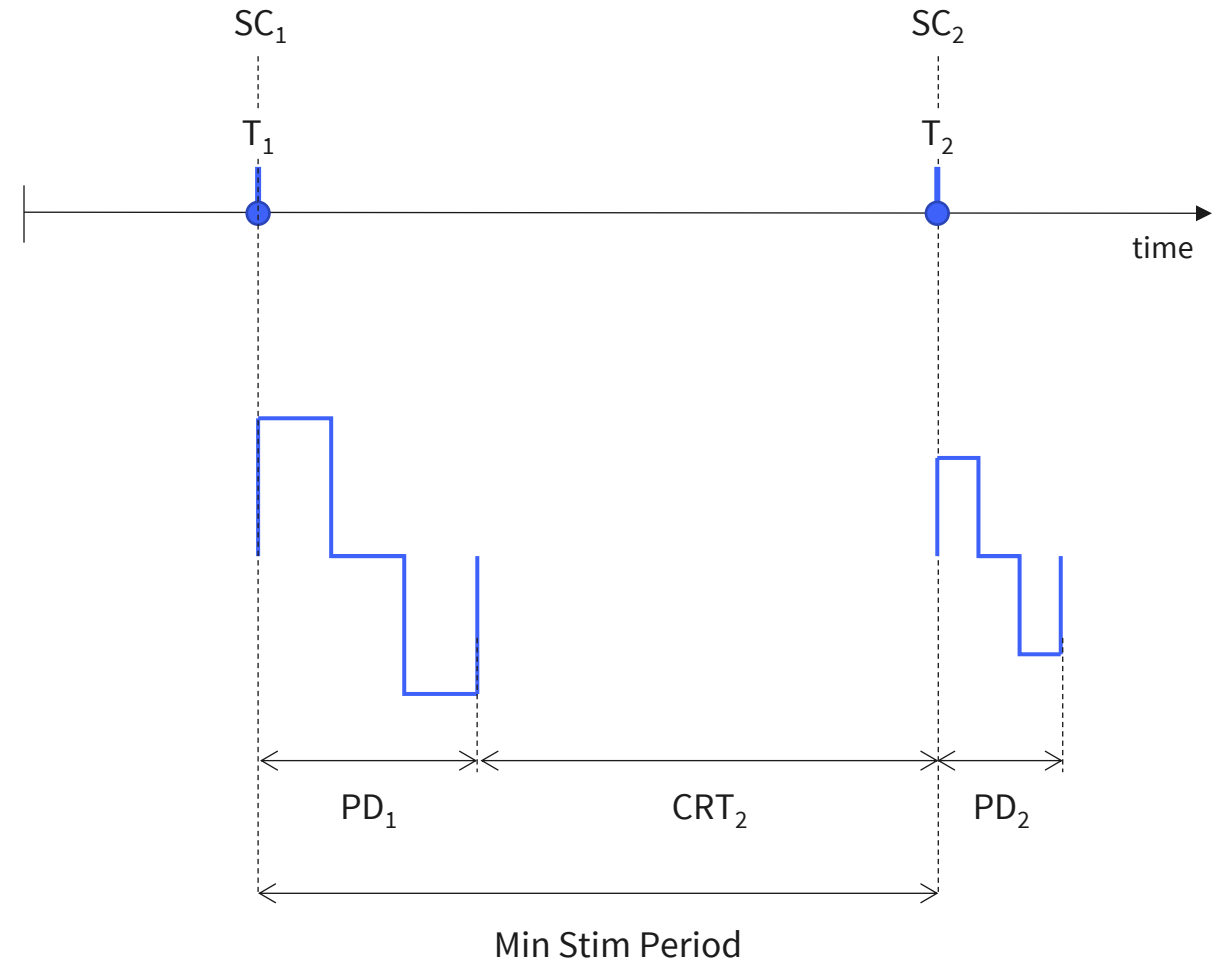
Stimulation Capabilities - 2

Frequency

- The minimum distance between two consecutive stimuli happening at T_1 and T_2 (Min Stim Period) is the sum of the following two values rounded up to the nearest stimulator time resolution increment (time resolution = $10\mu\text{s}$):
 - The Pulse Duration of the first stimulus (PD_1)
 - The Chip Reconfig Time for the second stimulus (CRT_2), i.e. the time needed to reconfigure the chip based on the SC_2 (note that only SC_2 is relevant and not SC_1)
- Chip Reconfig Time = 0 if $SC_2 = SC_1$, i.e. when the spatial configuration does not change. In this scenario very high stimulation frequencies can be achieved (e.g. 4kHz for a Pulse Duration of $250\mu\text{s}$).
- If $SC_2 \neq SC_1$, The Chip Reconfig Time depends from the number of independent rows (N_{ROWS}) required by SC_2 , that is the union of the rows in the electrode array when considering all endpoints for SC_2 , according to the formula:

$$\text{Chip Reconfig Time} = 26\mu\text{s} + 8.4\mu\text{s} * (N_{\text{ROWS}} - 1)$$

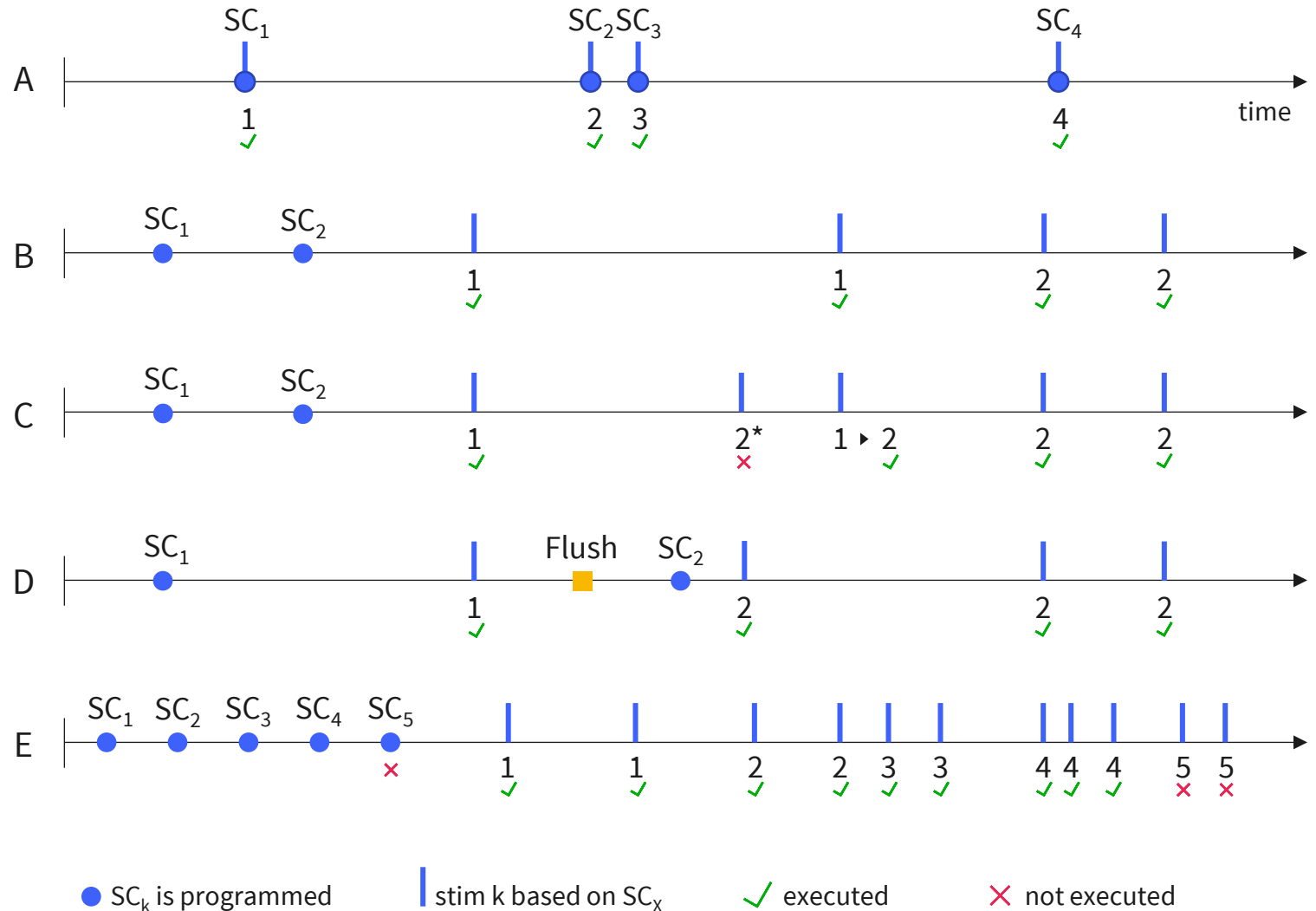
For instance, if $PD_1 = 150\mu\text{s}$ and SC_2 has 2 endpoints on 2 different rows ($N_{\text{ROWS}} = 2$), Chip Reconfig Time = $150 + 26 + 8.4\mu\text{s} = 184.4\mu\text{s} \Rightarrow$ rounded up to $190\mu\text{s}$



Possibility of Programming Complex Stimulation Patterns

Examples and Constraints

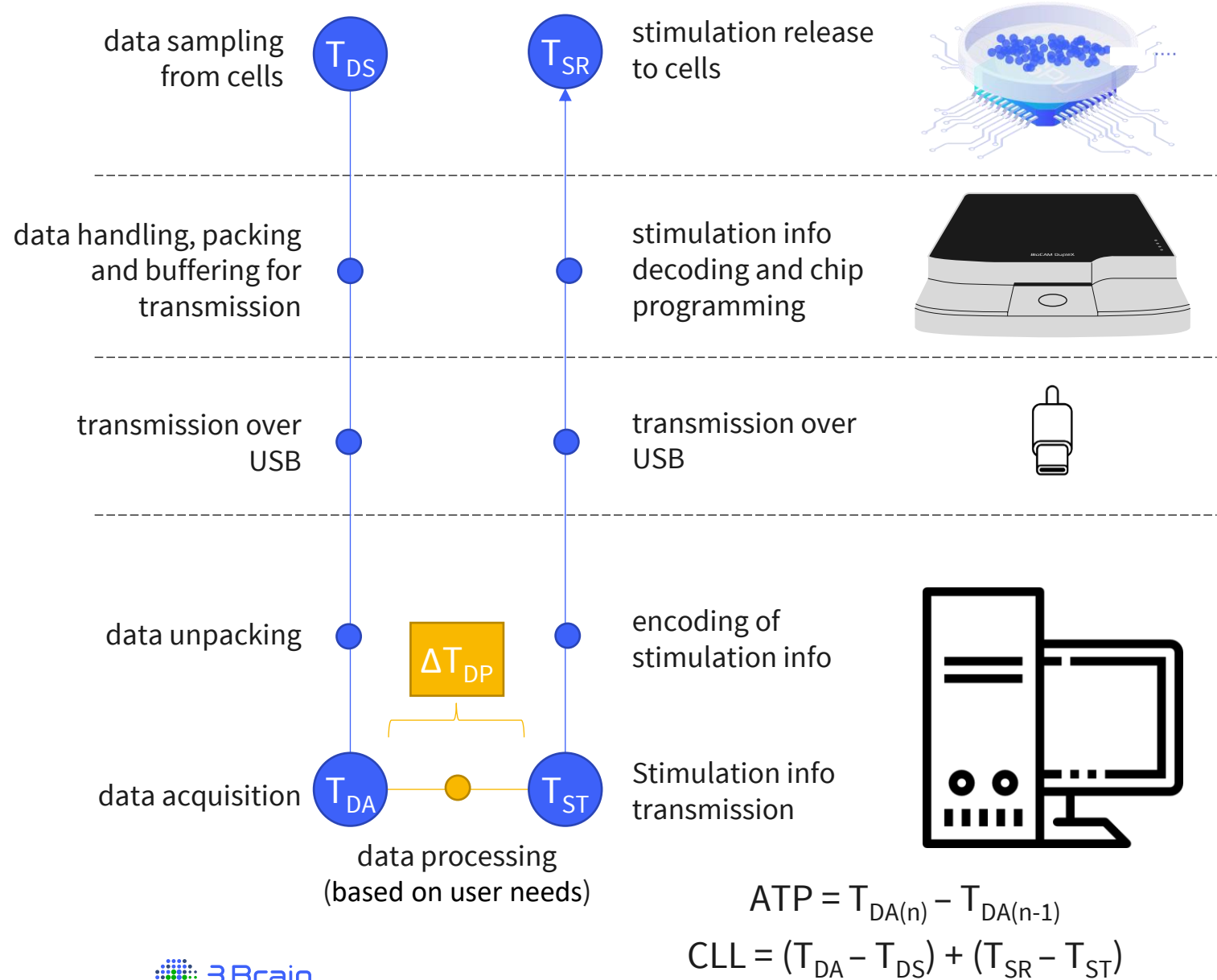
- A. Stimuli based on different SPs are released immediately and in succession, even at short intervals (e.g. 2 and 3; for specs see prev. slide).
- B. Stimuli are programmed to be released later on (up to 1k future stimuli). When programming stimuli must be temporally ordered.
- C. SC₂ tries to program three stimuli, however the first (*) is not executed (it's ahead the last from SC₁). Also, the last stimulus from SC₁ is converted to SC₂. (because SC₂ effectively takes place at (*) time).
- D. To overwrite future stimuli, the Flush command is first executed.
- E. Multiple SCs can be appended one after the other for future stimuli. The maximum number of SCs that can be buffered at a given time instant (SC_{max}) depends on how many electrodes are involved in those SCs. If all SCs use two electrodes SC_{max} = 50; regardless the number of electrodes used by SCs, it is guaranteed to have always SC_{max} ≥ 4.



Closed-loop applications

Method

- The API can be set to operate with a custom Acquisition Time Period (ATP, i.e. a fixed amount of data that is acquired at each acquisition cycle) between 1ms and 250ms.
- We measure the Closed-Loop Latency (CLL) as the total time between the data sampling from cells (T_{DS}) and the stimulation release (T_{SR}), i.e. the time to close the loop, without the time needed to process the data (ΔT_{DP}), which is based on user needs. The CLL measure is very accurate and it is done by timestamping (accuracy ~10ns) both sampled data and stimulation information.



Performances for closed-loop applications

Considerations

- Windows is not a real-time OS, therefore the measured CLL can experience fluctuations subject to several factors, such as processor clock speed and processor load.
- CLL is also affected by the number of EndPoints selected for a given stimulation, which affects the time to send instructions over USB and programming the chip.

Acq. Time Period [ms]	# of Total EndPoints	Closed-Loop Latency [ms]	
		Mean $\pm$ SD	Max
1	2	1.15 $\pm$ 0.04	1.52
2	2	2.18 $\pm$ 0.04	2.60
3	2	3.21 $\pm$ 0.04	4.41
5	2	5.20 $\pm$ 0.04	6.12
5	100	5.18 $\pm$ 0.04	6.20
10	2	10.25 $\pm$ 0.04	12.10
10	100	10.26 $\pm$ 0.04	11.47

Tests done on AMD Ryzen 9 7950X (4.50GHz) on 04 September 2023, fw v.1.7.2.4. For each row, the values were calculated from a sample of 11k CLL measurements.